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Dynamic excitation/inhibition balance preceding seizure onset and its link to functional and structural brain architecture

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Abstract

Background Altered excitation/inhibition (E/I) balance is a key mechanism in epilepsy, but its dynamic changes before seizure onset remain unclear. The interictal suppression hypothesis suggests that inhibitory input isolates epileptic regions, yet little is known about E/I dynamics in the pre-ictal phase.

Methods We analyzed high-density EEG recordings from patients with drug-resistant focal epilepsy, each with at least one recorded seizure. Cortical activity was reconstructed using source modeling, and time-resolved changes in the aperiodic exponent—a non-invasive marker of E/I balance—were computed. Directed functional connectivity was assessed via spectral Granger causality. These metrics were correlated with cortical thickness and neurotransmitter receptor density maps.

Results The aperiodic exponent increased progressively in the minutes preceding seizures, reflecting a global shift toward cortical inhibition, with no significant differences between epileptic and non-epileptic regions. Delta and theta power also increased preictally. Epileptic regions showed significantly more outward than inward connectivity, and more outward connectivity than non-epileptic areas. In non-epileptic regions, higher inhibition was associated with greater outward connectivity. Cortical thickness positively correlated with inhibition only in non-epileptic areas. Lower muscarinic receptor density was associated with stronger inhibitory shifts.

Conclusions Seizure onset is preceded by a widespread shift toward inhibition, possibly representing a compensatory mechanism that fails as ictal thresholds are crossed. This shift is linked to altered connectivity and disrupted structure–function coupling in epileptic regions. The dynamic aperiodic exponent emerges as a promising biomarker for seizure prediction and a potential target for neuromodulation strategies.

Keywords Excitation/inhibition balance, Spectral aperiodic component, Seizure onset, Focal epilepsy, Network dynamics

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Background

The healthy brain maintains a dynamic balance between excitatory and inhibitory (E/I) signals, enabling efficient information transfer across cortical networks [1, 2]. Disruptions to this balance represent one of the leading mechanisms in several neurological disorders [3, 4]. Epilepsy is associated with E/I imbalance which facilitates abnormal neuronal synchronization, ultimately leading to seizures [5, 6].

Multiple methods have been developed to estimate E/I balance using electroencephalography (EEG) [7]. Among these, the aperiodic exponent of the EEG power spectrum has emerged as a sensitive marker of cortical E/I balance, allowing full-brain assessments in clinical populations [8–13]. The aperiodic component is subject to certain limitations as a proxy of E/I balance. Previous work emphasized that, although numerous indices of E/I balance derived from resting EEG activity have been proposed, their validation remains insufficient, suggesting some methodological pitfalls in the interpretability [14, 15]. However, recent studies have strengthened the link between aperiodic and E/I balance: pharmacological manipulations targeting GABA_A receptors have been shown to increase the aperiodic exponent, consistent with enhanced cortical inhibition [16], while flatter slopes have been linked to higher glutamatergic tone, reflecting increased excitatory drive [17]. Furthermore, the aperiodic exponent has been increasingly applied in epilepsy research as both an E/I marker and a feature for seizure prediction, as highlighted in a recent systematic review [18]. Taken together, these findings suggest that, despite its methodological limitations, the aperiodic exponent may represent a robust and practical proxy for assessing E/I balance in both health and disease. Recent findings using this measure in epilepsy revealed an inhibition increase in epileptic regions during the interictal phase, suggesting links between E/I dynamics, treatment response, and cognitive functioning [19]. Additional evidence showed that interictal spikes are often accompanied by increased inhibitory activity in distant brain regions. This widespread inhibition may act as a protective mechanism, mitigating the impact of epileptiform discharges and supporting cognitive performance in drug-resistant epilepsy [20]. These findings align with the “interictal suppression hypothesis” (ISH), which posits that epileptic zones receive stronger inward inhibitory inputs from surrounding brain regions, functionally isolating them and limiting propagation of pathological activity [21]. However, while most evidence supports this framework in the interictal state, the transition from interictal to ictal dynamics remains less well understood. Seizure generation is a dynamic process involving gradual shifts in E/I balance, pushing the system from an

interictal to an ictal state bifurcation [22]. Computational and empirical findings suggest that changes in E/I can reorganize network influence, amplifying or attenuating inter-regional communication [23]. Understanding E/I dynamics preceding a seizure and the relationship with whole-brain functional organization is crucial to clarifying neural mechanisms underlying seizure initiation [24]. To this purpose, we investigated pre-ictal E/I balance and directional functional interactions (inward/outward connectivity) using dynamic estimates of the EEG aperiodic exponent and spectral Granger causality. We analyzed high-density EEG data from a cohort of drug-resistant epilepsy patients, reconstructing cortical source activity across interictal and pre-ictal epochs [25].

We hypothesized that the slope of the power spectrum density (PSD) aperiodic component would increase linearly as seizure onset approached, reflecting a dynamic change in E/I balance leading to the ictal event [26]. We also expected that dynamic E/I change preceding seizures would have downstream effects on the brain’s functional architecture, namely the way in which regions interact across the brain. Relevantly, recent findings suggested that E/I dynamics and brain functional organization are shaped by micro-structural properties both at the local and global scales [27, 28]. At the micro-scale, seizure generation is strongly influenced by neurotransmitter receptor density. For instance, increased expression of excitatory glutamatergic receptors (AMPA, NMDA) or reduced density of inhibitory GABA_A receptors alters local excitatory/inhibitory balance. These microscopic changes shift the threshold for synchronized firing, predisposing networks to pre-ictal instability. Similarly, brain morphology regulates how local circuits integrate inputs and therefore how activity flows into the brain. Importantly, previous findings have shown that EEG aperiodic activity and functional organization of intrinsic brain activity, in patients with epilepsy, correlates with cortical expression of genes involved in E/I regulation as well as with morphological cortical organization [29]. Therefore, these microscopic processes can also be reflected in EEG recordings of pre-ictal activity, where disruption of excitation/inhibition balance could manifest as abnormal and dynamic changes of EEG aperiodic component. In this way, micro-scale brain features such as, brain morphology and neurotransmitter receptor distributions, scale up to influence pre-ictal EEG dynamics, ultimately driving the transition to seizure. In this light, to examine these interactions between brain architecture and function in relation to seizure onset, we specifically assessed how cortical thickness and neurotransmitter receptor densities relate to pre-ictal E/I dynamics.

Overall, this study aims to advance our understanding of large-scale brain organization related to seizure

initiation by integrating dynamic EEG measures with structural and molecular brain features, leveraging non-invasive high-density EEG methods.

Methods

Participants

Patients with drug-resistant focal epilepsy underwent prolonged high-density EEG for clinical pre-surgical evaluation between 2021 and 2024 at the Epilepsy and Clinical Neurophysiology Unit, IRCCS Eugenio Medea in Conegliano (Italy). We included patients with at least one seizure recorded with high density EEG. The presurgical workflow included Video EEG monitoring (32 channels, 4 days), 24 h with high density EEG, brain magnetic resonance imaging (MRI), and positron emission tomography as an adjunctive investigation in selected patients. Twenty patients were retrospectively screened and included in the sample [mean age=35.50 (SD=16.94); 9 females; 9 TLE; 6 frontal epilepsy; 4 parietal epilepsy; 1 unknown]. 85% of patients were drug-resistant. Patients’ demographic and clinical characteristics are provided in Table 1. Patients inclusion criteria: (i) age of 16 and 75 years with focal epilepsy; (ii) admitted to the epilepsy monitoring unit (EMU) for phase I non-invasive pre-surgical investigation; (iii) detection of electrographic or electroclinical epileptic seizures during high-density EEG. Diagnostic confirmation of epilepsy and epilepsy type, as well as drug-resistance, was according to ILAE guidelines [30]. Diagnostic confirmation relied on the combined information acquired during phase I of the presurgical assessment, relying on clinical presentation, clinical semiology, MRI findings, EEGs, positron

emission tomography (PET) results. Seizures were identified and classified according to the American Clinical Neurophysiology Society (ACNS) 2021 definitions [31]. Exclusion criteria: (i) vascular or non-low-grade tumors as causes of epilepsy; (ii) large artifacts on MRI (it would result in a less precise segmentation and source imaging); (iii) multifocal patients, based on phase I of the assessment. The study protocol was conducted according to the Declaration of Helsinki and approved by the local ethical committee (Comitato Etico Area Nord Veneto). All participants provided written informed consent to participate in the study. The methodology followed the STROBE guidelines for a cohort study.

EEG recording and preprocessing

High-density EEG was recorded with a 128-channel Micromed System Plus device. The signal was sampled at 512 Hz and referenced to the vertex. The impedance of the electrodes was kept <5 kΩ. Seizure onset was marked by two epileptologists with expertise in EEG (AD and PB). For each subject, we collected 10 min of interictal (resting state) high-density electroencephalography (HDEEG) activity during wakefulness, in a separate recording with an average distance from the seizure of at least 3 h (range 3–5 h). Signal preprocessing was conducted using EEGLAB version 14.1.2b. The preprocessing steps included (a) data resampling to 250 Hz; (b) band-pass filter (0.1–45 Hz) with Hamming window—finite impulse response filter; (c) 1-s data epoching; (d) rejection of epochs and channels with artifacts; (e) average reference; (f) bad channels interpolation (spherical spline method); (g) independent component analysis (ICA) (400 components-Infomax algorithm); (h) automatic artifact-related ICA components removal; and (i) concatenating the epochs to reconstruct a continuous signal.

The automatic artifact and bad-channel detection was performed using the Trial-by-Trial plugin in EEGLAB with the following parameters: channels were marked as bad if their differential average amplitude exceeded 250 μV in more than 30% of the epochs. Epochs were excluded if they contained more than 10 bad channels. Flat channels were detected and removed using the Trimoutlier plugin, with a lower threshold of 1 μV. For the automatic ICA rejection, we used the ICLabel plug-in [32]. The ICA components classified as brain were kept (after visual inspection), while components classified as eyes, muscle, channel or line noise, heart or others within a 70–100% accuracy range were removed.

Data

Duration of pre-ictal EEG data ranged from 2 to 160 min. On average, 2.93 ± 3.75 (standard deviation; SD) channels were interpolated, 432.80 ± 504.21 epochs and

Table 1 Sample demographic and clinical information. Demographic and clinical features of the patients. MRI findings are reported based on neuroradiological examination

Patients	Mean ± standard deviation
Age	35.50 ± 16.94
Age of onset	20.80 ± 16.65
Number of antiseizure medications	2.05 ± 1.16
Duration of epilepsy (years)	14.73 ± 14.90
MRI	Number
Cavernous angioma	1
Cerebellar atrophy	1
FCD	9
HS	3
Polymicrogyria	1
MRI-negative	4
Unknown	1

Abbreviation of MRI abnormalities: FCD focal cortical dysplasia, HS hippocampal sclerosis

12.48 ± 4.66 components were rejected. After excluding two patients for whom pre-ictal data lasted less than 5 min, the minimum pre-ictal data duration after the cleaning was 13 min. In order to be consistent across participants, we selected for each subject 13 min of signal prior to the seizure for a total of 29 analyzed seizures recorded with HDEEG (average 1.35; range, 1–3). For the interictal activity, the preprocessing resulted on average into, 0.11 ± 0.48 (mean $\pm$ SD) channels, 44.64 ± 9.51 epochs and 12.35 ± 4.60 components rejected. After preprocessing, each subject had at least 7.5 min of artifact-free continuous signal. To harmonize data across subjects, we selected 450 s for the interictal (resting state) data of each participant.

Cortical source modeling

Individual MRIs used for source imaging consisted of high-resolution T1-weighted three-dimensional (3D) isotropic (1 mm) acquisitions, collected at Ospedale San Bortolo, Vicenza, with a Siemens Skyra scanner 3 T. The images were segmented using the Freesurfer (7.3.2) standard pipeline with default parameters [33]. Brain surfaces and Freesurfer-derived cortical thickness maps were then imported into Brainstorm [34]. Electrode co-registration with MRI anatomy was carried out in Brainstorm using standard anatomical landmarks, with manual adjustments applied if necessary. The cortical surface mesh was downsampled to 15,002 vertices. Three boundary element model (BEM) surfaces—inner skull, outer skull, and scalp—were reconstructed (1082 vertices per surface). The forward model was computed using a three-shell BEM approach (default Brainstorm parameters) using Brainstorm integrated OpenMEEG plugin [35]. For the inverse solution, we employed the weighted minimum norm estimate, using Brainstorm's default parameter [36] settings. Source activity was then downsampled to 68 cortical regions of interest (ROIs) according to the Desikan-Killiany [37] atlas. ROI time series were obtained as the mean time series across ROI vertices. The same atlas mask was applied to cortical thickness maps to obtain a thickness value per ROI. Finally, for each subject, we identified the presumed seizure onset zone, from here after defined for simplicity as epileptic regions, in the following way: (i) post-surgical outcome; (ii) information available at Phase I, specifically clinical semiology, MRI imaging, EEG pattern, and ancillary clinical information; and (iii) source estimation of ictal activity (where possible) or interictal epileptiform discharges (IEDs) following methodology of previous works of the group [25, 38]. All information was validated by two independent epileptologists (PB, AD). To assign epileptic regions to a Desikan-Killiany parcel, we adopted a methodology used in our previous work [38]. Namely, if

the epileptic regions were placed at the border between or spread across two atlas parcels, despite the size of the spread, we included both of the regions in the classification of the epileptic focus. This approach can be justified by the following reasons: (i) The primary goal of defining the epileptic regions is to ensure that no potentially relevant regions are missed; (ii) parcels at the margins of the epileptic regions may represent transition zones between the core epileptic tissue and surrounding regions, including them provides a more comprehensive representation of the network involved; and (iii) an inclusive approach minimizes the risk of false negatives due to partial volume effects, signal leakage, or border overlap. The atlas parcels of the epileptic regions were identified on the individual cortex, reconstructed from patients MRI segmented with Freesurfer. By contrast, all remaining cortical parcels were classified as non-epileptic regions.

Dynamic parameterization of aperiodic and periodic activity

Source space time-series from ROIs were processed with the SPRINT Brainstorm plug-in [39]. In brief, SPRiNT builds on the specparam algorithm [8] by applying it within a sliding-window framework based on the short-time Fourier transform (STFT). For each time window, the neural time series is transformed into its power spectral density (PSD), which is then decomposed by specparam into aperiodic and periodic components. Specifically, specparam first estimates the aperiodic background activity by fitting a $1/f$ -like function to the PSD in log–log space, modeling it as an offset and exponent term. Once the aperiodic component is fit and subtracted, residual peaks above this background are identified as periodic (oscillatory) components. The aperiodic exponent is thus directly obtained from the slope of the log–log PSD fit in each window, providing a time-resolved measure of scale-free activity. We used the following setting parameters: window length for short time Fourier transform = 60 s; overlap = 90%; frequency range 1–40 Hz; averaged FFT per time point = 1. The parameters of the specparam algorithm were set as follows: frequency range 1–40 Hz; peak width limits = [0.5, 12]; maximum number of peaks = 3; minimum peak height = 3 dB; peak threshold = 2; aperiodic mode = “fixed”; guess weight = “none.” SPRiNT thus yielded time series of both aperiodic (i.e., $1/f$ exponent) and periodic signals (i.e., oscillatory power), resulting in 120 values per ROI in the pre-ictal period and 66 in the resting state. SPRiNT results also provided a time–frequency map for each ROI, allowing to investigate dynamic changes of the power spectrum. We grouped oscillatory results into the 5 canonical frequency bands:

Delta (2–4 Hz), Theta (5–7 Hz), Alpha (8–12 Hz), Beta (13–20 Hz), Gamma (21–40 Hz).

Inward/outward functional connections

Effective functional connectivity was computed using spectral Granger causality (GC) via fieldtrip toolbox [40] with a model order of 10. To track dynamical changes of effective connectivity, we used a sliding window approach with the same frequency range, window length and overlap of SPRiNT in order to obtain the exact same time resolution of dynamic GC and 1/f exponent. We then grouped GC results into the 5 canonical frequency bands previously described. We applied a proportional threshold to the connectivity matrix keeping only connections above the 95th percentile, in line with previous evidence showing that this method ensures equal density of the brain networks across subjects [41–43].

Neurotransmitters receptor density maps

Neurotransmitter receptor density maps were extracted by the seminal work of Hansen and colleagues [27]. Briefly, receptor densities were estimated using PET tracer studies for a total of 18 receptors and transporters, across 9 neurotransmitter systems. Details of each study, the associated receptor/transporter, tracer, number of healthy participants, age, and reference with full methodological details can be found in Table 1 of Hansen et al. [27]. A more extensive table can be found in the supplementary material of Hansen et al. [27] (Table S3) which additionally includes the PET camera, number of males and females, PET modeling method, reference region, scan length, modeling notes, and additional references, if applicable. In all cases, only healthy participants were scanned (total $N=1238$, 718 males, 520 females). Altogether, the images are an estimate proportional to receptor density and we therefore refer to the measured value (i.e., binding potential, tracer distribution volume) simply as density. Volumetric PET images were registered to the MNI-ICBM 152 nonlinear 2009 (version c, asymmetric) template, averaged across participants within each study, then parcellated to 68 cortical regions according to the Desikan Killiany atlas. In this study, we extracted the parcellated neurotransmitters receptor density maps of the following neurotransmitters, as mainly related to E/I balance and seizure generation: GABA (GABA_A) [44], acetylcholine (VACHT, M1, a4b2) [45], glutamate (NMDA, mGluR5) [46], opioid (MOR) [47], and serotonin (5-HTT, 5-HT1a, 5-HT2a, 5-HT1b, 5-HT4, 5-HT6) [48]. For information about parcellated neurotransmitters receptor density brain maps see Data availability section.

Statistical analysis

For each time series, namely aperiodic exponent and power spectrum density both in pre-ictal and resting state condition, we applied a linear regression approach to extract regression slopes. To test the differences between slopes, we applied two sets of contrasts: within conditions (pre-ictal and RS) contrasting epileptic vs. non-epileptic regions; between conditions: contrasting epileptic in pre-ictal vs. resting state (RS) and non-epileptic in pre-ictal vs. RS. Specifically, from each regression we extracted the slope coefficients (β) and coefficient standard error (SE). To compare the difference of the two slopes, we applied the following formula: $\beta_1 - \beta_2 / \sqrt{SE(\beta_1)^2 + SE(\beta_2)^2}$. The p values were extracted from the t-distribution.

Successively, correlations between aperiodic exponent and GC-derived inward and outward connectivity measures, as well as between aperiodic exponent and cortical thickness were performed using Spearman's correlation with false discovery rate (FDR) correction for multiple comparisons. Finally, a spin-permutation approach was applied to investigate the relationship between the cortical distribution of pre-ictal aperiodic exponent and the neurotransmitter density maps while controlling for the effects of spatial autocorrelation [49]. More specifically, the empirical Spearman's correlation between each receptor density brain map and the pre-ictal aperiodic exponent brain map were correlated. Next, a null distribution of 5000 null correlations were generated by randomly spatially rotating the receptor density brain maps 5000 times and correlating these spatial autocorrelation-preserving null brain maps with the empirical aperiodic exponent map ("spin test"; [50]). Statistical significance was calculated as the frequency with which the empirical correlation coefficient was greater than the absolute null correlation coefficients. This conservative statistical approach ensures the reported correlations are not driven by the inherent spatial autocorrelation of brain maps (Fig. 1).

Results

E/I balance dynamics preceding seizure onset

Whole-brain dynamics are characterized by an increase in the aperiodic exponent as the seizure onset approaches, suggesting a global shift in the excitation/inhibition (E/I) balance toward inhibition (Fig. 2A). Pre-ictal slope of the aperiodic component was steeper as compared to the interictal period, for both the epileptic and non-epileptic regions (epileptic regions pre-ictal vs. interictal: $t=2.032$; $p=0.046$; non-epileptic regions pre-ictal vs. interictal: $t=6.94$; $p<0.001$). Notably, no significant slope differences were observed

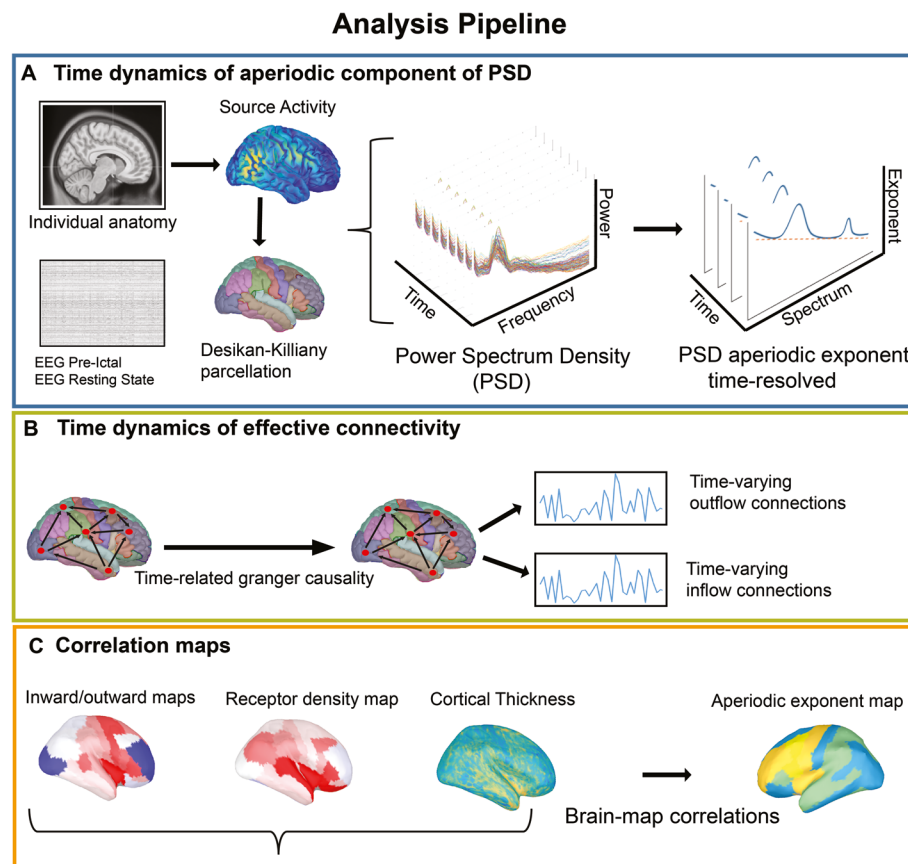


Fig. 1 Analytical pipeline. **A** Electrical source imaging of the pre-ictal and interictal data; Desikan-Killiany atlas activity downsample and computation of dynamic changes of aperiodic component across time. **B** Time-dynamic of effective connectivity (spectral Granger causality) to successively obtain time-varying inflow and outflow connections. **C** Correlation between the exponent of spectral aperiodic component with: in/outward connections, receptor density maps and cortical thickness

between epileptic and non-epileptic regions during the pre-ictal phase ($t = 1.38$; $p = 0.16$; see Fig. 2A), suggesting that E/I dynamic changes extend beyond the epileptic network.

To provide evidence of the consistency of our results, we show results of the aperiodic exponent dynamics in larger pre-ictal time windows, 20, 30, 40 min, respectively. As expected, increasing the window size led to a linear decrease in the number of available subjects: 24 subjects for the 20-min window, 16 for the 30-min window, and 11 for the 40-min window. Beyond 40 min, the reduction in sample size rendered the results unreliable; therefore, we limited our analysis to this range. The time-series plot highlights how the large exponent increase is expressed in the last part of the pre-ictal period (see Additional File 1: Fig. S1). These results support that the pre-ictal selected time window (13 min before seizures), representing a good balance between signal length, number of patients included in the analysis and the characterization of aperiodic exponent dynamics.

Similar findings were observed for the periodic component (power spectrum). Specifically, delta and theta frequency bands were characterized by a power increase as the seizure approached (see Fig. 2B). This result was supported by the difference in the slope of the PSD in time, when comparing pre-ictal to an interictal (resting state) time intervals (see Table 2). No differences were identified when comparing the slopes epileptic vs. non-epileptic regions in delta and in theta bands ($p > 0.05$).

Finally, we correlated the slope of the pre-ictal aperiodic component increase with the number of antiseizure medications (ASMs). We did not find any significant relationship between these two variables ($\rho = 0.278$; $p = 0.145$).

Inward/outward connections and relationship with exponent

When investigating time-dependent changes of in/outward connections in delta and theta bands in the pre-ictal phase, we did not find any significant linear increase

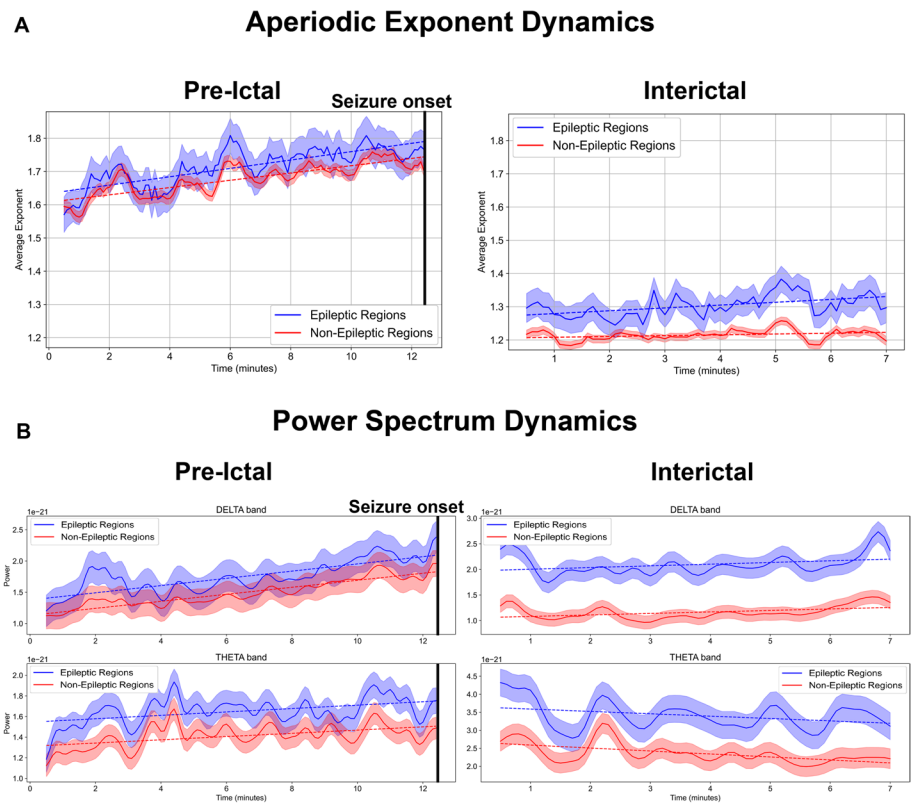


Fig. 2 Aperiodic and periodic dynamics approaching a seizure and in resting-state. **A** Dynamic changes in the aperiodic exponent. Top-left panel, an increase in the aperiodic exponent can be observed as the seizure approaches. Conversely, the top-right panel shows the aperiodic exponent during the interictal state, where no clear trend is detectable. **B** shows the temporal dynamics of delta and theta power (periodic component for pre-ictal and interictal conditions). Delta and theta power increases as the seizure approaches. For both panels, blue lines indicate the behavior in epileptic regions, and red lines correspond to non-epileptic regions. The shaded area around the lines represents the standard error. The black line marks the seizure onset

Table 2 Slope analysis results. The table show the significant contrast of the slope analysis contrasting the dynamical changes in oscillatory activity (i.e., delta and theta bands)

Contrast	t value	p value
Epi ROI pre-ictal vs. RS Delta	1.88	.064
Non-Epi ROI pre-ictal vs. RS Delta	3.44	.001
Epi ROI pre-ictal vs. RS Theta	3.50	< .001
Non-Epi ROI pre-ictal vs. RS Theta	6.50	< .001

or decrease as the seizure approached (see Fig. 3A). Consequently, we shifted our focus to time-averaged inward and outward connections. Specifically, we averaged over time inward and outward connections of the pre-ictal period (13 min prior to seizure), separately. The same procedure was performed for the interictal period. We then compared connectivity profiles between epileptic and non-epileptic regions within pre-ictal and interictal phases, separately. Successively, in order to examine the

relationship between E/I balance and functional organization (in/outward connectivity), we correlated time averaged aperiodic exponent (pre-ictal and interictal) with directed connectivity previously averaged over time.

Granger causality analysis in the delta band during the pre-ictal phase revealed a significantly greater number of outward connections compared to inward connections within epileptic regions ($t=2.12, p=0.042$). Furthermore, epileptic regions exhibited a greater number of outward connections compared to non-epileptic regions ($t=2.29; p=0.030$) (see Fig. 3B). Similar results were found for the theta band, and are reported in the Supplementary Materials (SM) (see Additional File 1 Fig. S2A, B). By contrast, no significant differences were detected when comparing inward vs. outward connectivity within and between epileptic and non-epileptic regions.

As per the relationship between E/I and effective connectivity, larger inhibition (i.e., higher exponent values) were related to an increased number of outward connections in the non-epileptic regions in the pre-ictal phase ($\rho=0.48; p=0.044$). Interestingly, in the pre-ictal

Delta band connectivity

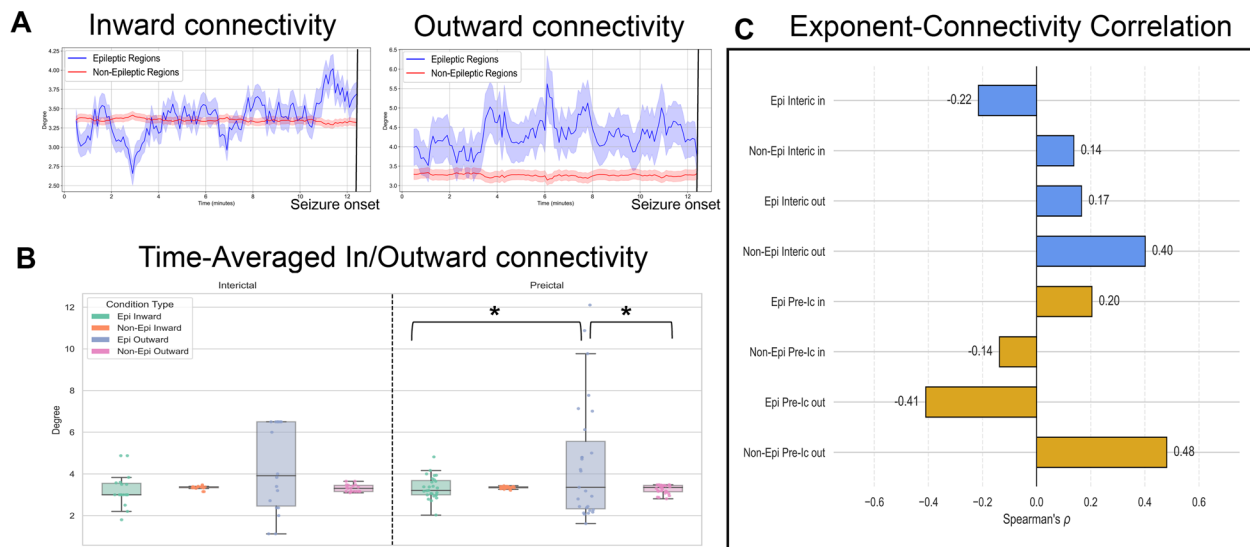


Fig. 3 Effective connectivity in the delta band and its relationship with the aperiodic exponent. **A** Dynamics of the inward and outward number of connections (degree) in the delta band, for epileptic (blue lines) and non-epileptic (red lines) regions. Shaded areas indicate the standard error, and black vertical lines mark the seizure onset. **B** Distribution of the time-averaged degree during interictal and pre-ictal conditions using box plots. Significant comparisons are indicated by black lines with asterisks. **C** Spearman's rho values for the correlation between the aperiodic exponent and the delta band inward (in) and outward (out) connectivity, represented as a barplot. Results are shown separately for epileptic (Epi) and non-epileptic (Non-Epi) regions, as well as for interictal (Interic) and pre-ictal (Pre-Ic) intervals

period, epileptic regions displayed an opposite correlation pattern as compared to the non-epileptic regions, with more outflow connections as the E/I balance shifted toward excitation. Even though the latter correlation does not survive multiple comparison correction ($\rho = -0.41$; $p = 0.067$), it is worth mentioning the pattern inversion in the granger-exponent correlation between epileptic and non-epileptic regions in the pre-ictal period. On the other hand, both epileptic and non-epileptic, in the interictal phase, display the same correlation pattern between outflow connections and the aperiodic exponent, with no significant effect surviving multiple comparisons correction (see Fig. 3C). Similar behavior was detected in the results related to the theta band (see Additional File 1: Fig. S2C).

Exponent-cortical thickness/receptors correlation

To investigate dynamic changes of structure–function coupling approaching the seizure, we performed a correlation analysis between cortical thickness and aperiodic exponent in the pre-ictal interval, in each time point, separately for epileptic and non-epileptic regions. We observed a stable correlation over time, suggesting non-significant changes in the structure–function coupling as the seizure approached (see

Fig. 4A). For this reason, we focused on the correlation between time-averaged value of the exponent in the pre-ictal period and the cortical thickness.

We identified a distinctive pattern between epileptic and non-epileptic regions. Specifically, larger values of cortical thickness were associated with increased inhibition (i.e., larger exponent values) ($\rho = 0.21$; $p \leq 0.001$), whereas no correlation was detected in epileptic regions ($\rho = -0.04$; $p = 0.61$; see Fig. 4A).

Finally, E/I balance (i.e., aperiodic exponent) in the pre-ictal period was significantly negatively correlated with muscarinic receptor density ($\rho = -0.433$; $p = 0.013$; see Fig. 4B). This suggests that an E/I balance is shifted towards inhibition in regions with lower muscarinic receptor density.

Discussion

This study provides novel insight into the dynamic modulation of excitation/inhibition (E/I) balance in the human cortex preceding seizure onset, with a particular focus on its relationship to large-scale network interactions, cortical structure, and neurotransmitter receptor density. The findings suggest that seizure initiation may be preceded by a global, rather than strictly focal, reorganization of cortical excitability.

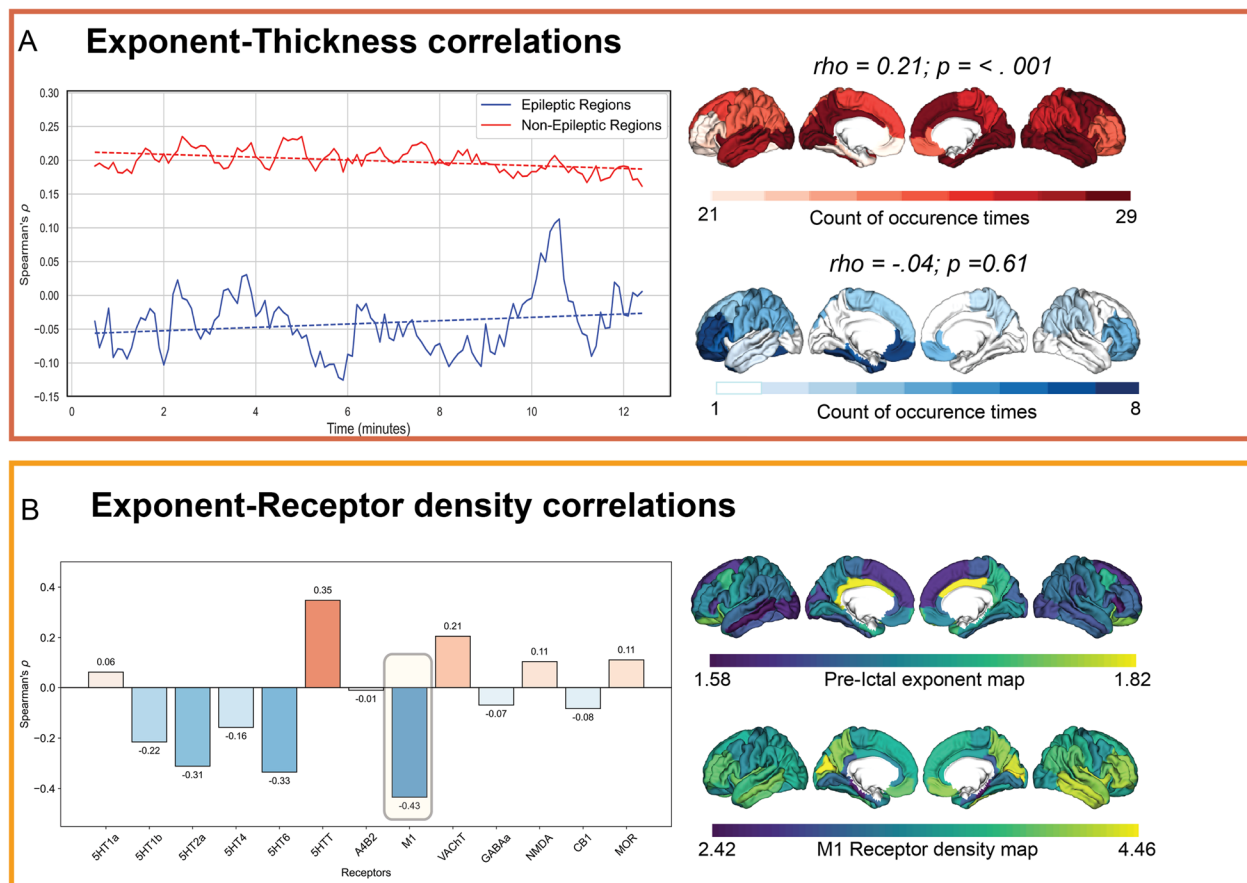


Fig. 4 Correlational results. **A** In the top-left panel we show time-dependent connectivity between pre-ictal aperiodic exponent and the cortical thickness for epileptic (blue line) and non-epileptic (red line) regions. Brain plots on the top-right represent the counts (number of subjects) in which that specific region is considered non-epileptic (red regions) or epileptic (blue regions). **B** Spearman's rho derived from spin-permutations between receptor density of multiple neurotransmitters and the brain map of the exponent of spectral aperiodic component averaged over time. The yellow shaded area highlights the correlation which survived FDR correction. Spatial distribution on the cortex (brain map) of the aperiodic exponent and the muscarinic receptor density are represented in the lower-right panel

Global inhibitory shift precedes seizure onset

In the present work, by leveraging electrical source imaging, we showed that the aperiodic exponent, a non-invasive marker of E/I balance, dynamically increases as the seizure approaches. This effect was observed across both epileptic and non-epileptic regions, suggesting that seizure initiation may be preceded by a global reorganization of cortical excitability rather than a strictly focal process. Moreover, by comparing pre-ictal activity with an interictal condition, we showed the specificity of this dynamic pattern for the pre-ictal period.

It is important to note that while the aperiodic exponent has often been linked to excitation/inhibition (E/I) balance, its neurobiological underpinnings are likely multifactorial. Previous work suggests that the slope reflects not only inhibitory tone, but also synaptic filtering properties and the sparsity of network activity [9]. Therefore, the increase we observed may not unambiguously

indicate “inhibition” per se, but more broadly a shift in cortical population dynamics that biases the system toward reduced excitability. Within this broader framework, our interpretation of a “global inhibitory shift” should be understood as reflecting cortical states dominated by inhibitory and/or filtering processes that limit network excitability.

Our results align well with recent intracranial EEG evidence suggesting a progressive inhibition in circadian rhythms of aperiodic exponent when approaching the seizure [26]. Our results extend previous findings by providing additional insight into the spatiotemporal dynamics of this process. Firstly, contrary to stereo electroencephalography (sEEG), our approach allows whole-brain sampling, and is not limited to the investigation of regions already known to be involved in seizure generation. Secondly, by integrating structural and receptor mapping, we provide additional insight into the

spatiotemporal dynamics of this process and into the biological substrates that may support it.

As previously mentioned, the hypotheses that we tested here were generated based on the interictal suppression framework—mainly tested with intracranial recording—posing that the epileptic regions receive net inward inhibitory connections as a result of protective mechanisms, in the interictal phase [21]. This is counter-intuitive based on the knowledge that on a cellular level converging evidence linked seizures to an hyperexcitable state [51]. In this light, the majority of current antiseizure medications (ASMs) work to restore E/I balance by either diminishing excitation or augmenting inhibition [52]. However, recent theoretical expansions are enlarging our understanding of the link between E/I balance and seizure generation, suggesting that hyperexcitability may not be a universal driver for seizure occurrence [53]. For example, dynamic assessment of the excitation/inhibition (E/I) balance in the rat hippocampus demonstrated that the initiation phase of seizure-like events (SLEs) is characterized by predominant inhibition in pyramidal neurons, with maximal synaptic excitability emerging during the ictal core of the SLE [6]. This temporal dissociation suggests that early inhibitory dominance may constitute a network-level homeostatic response. Accordingly, we propose that the global inhibitory shift observed in EEG reflects large-scale cortical dynamics aimed at maintaining stability by actively steering the system away from the bifurcation threshold leading to seizure onset. This global modulation may serve as a last-line compensatory mechanism to prevent seizure onset, which ultimately fails as the ictal threshold is surpassed. The concurrent increase in low-frequency (delta and theta) oscillatory power in the pre-ictal phase is also consistent with previous observations of cortical slowing before seizures [54, 55]. The simultaneous modulation of both aperiodic and periodic components may reflect coordinated, widespread changes in cortical excitability, suggesting that these spectral signatures may be functionally coupled.

Directionality of functional connectivity differentiates epileptic and non-epileptic regions

The effective connectivity analysis may help to better elucidate whole-brain organization prior to a seizure, and its relationship with the E/I balance. First, in line with previous evidence, our findings show that epileptic regions are characterized by a larger number of outflow connections as compared to the inward connections. Additionally, the outward connections number of epileptic regions is larger as compared to non-epileptic ones. These results align with previous intracranial findings, supporting the perturbatory role on the whole-brain network balance of the epileptic network [55]. We observed a positive

relationship between E/I balance and outward effective connectivity in non-epileptic regions. This association is specific to the period preceding seizure onset as opposed to the interictal period. These effects may be consistent with the protective inhibitory framework, whereby the brain attempts to isolate perturbatory nodes by reducing local connectivity and modulating the E/I balance through inhibitory influences, as suggested by recent sEEG findings [21, 56, 57].

This compensatory mechanism is mainly expressed in the slow frequencies, being potentially linked to network tuning processes at whole-brain scale. Low-frequency brain rhythms play a crucial role in coordinating neural activity across widely distributed brain regions, facilitating whole-brain network communication. Their long wavelengths allow them to propagate effectively over large distances, enabling the coordination of brain activity in spatially separated areas [58, 59]. Notably, transient changes in connectivity were less consistent than time-averaged measures, suggesting that structural organization and baseline functional architecture may exert dominant constraints on pre-ictal network dynamics.

Structural and molecular correlates of pre-ictal E/I balance

In assessing the anatomical and neurochemical underpinnings of E/I dynamics, we found that cortical thickness was positively correlated with the aperiodic exponent in non-epileptic areas. This is in line with literature linking thicker cortex to stronger inhibitory control, possibly due to a greater abundance of inhibitory interneurons or supporting glial architecture [60, 61]. By contrast, in the context of neurological disorders, cortical thinning is often associated with a loss of inhibitory capacity [62–65]. Notably, epilepsy has been associated with reduced cortical thickness, particularly within regions of the epileptic network [66, 67]. Relevantly, the structure–function coupling was absent in epileptic regions, which may reflect local pathological remodeling or dysmaturation that disrupts the normative cortical organization. The presence of the exponent-thickness correlation in the non-epileptic cortex, in the face of an approaching seizure, provides additional evidence for our proposed global inhibitory shift. It suggests that the non-epileptic regions of the brain are actively engaged in a protective compensatory mechanism, leveraging their structural integrity—thicker cortex associated with stronger inhibition—to steer the system away from the ictal threshold. The absence of this relationship within the seizure-generating network, which is often characterized by its own structural pathology and cortical thinning, may be a key factor in the local failure of this global protective response, ultimately leading to seizure onset. Our correlation findings between the exponent and cortical thickness strengthens

the argument that the observed E/I shift can be linked to an increase in inhibitory activity. However, our use of the term “inhibitory shift” should be interpreted within this broader context, signifying a collective change in neural population dynamics that is consistent with an increase in inhibitory influence, rather than being an exclusive and direct measure of synaptic inhibition. We believe this dynamic shift is a key component of the network’s compensatory response.

Furthermore, spatial correlations with neurotransmitter receptor density suggested a possible contribution of muscarinic cholinergic systems to pre-ictal E/I dynamics. Regions with lower muscarinic receptor density were associated with greater inhibitory shifts, potentially implicating cholinergic tone in the loss of cortical excitability balance prior to seizures. All in all, this result aligns with the important role of the cholinergic system in seizure generation [45, 68, 69]. This finding supports emerging evidence that neuromodulatory systems critically influence seizure susceptibility and offers a new avenue for targeted pharmacological interventions.

Potential applications and limitations

One of the most promising implications of this work is that the observed dynamics of E/I balance may represent a potential target for predictive biomarkers of seizure onset. The spectral aperiodic exponent provides a computationally efficient and non-invasive measure of cortical population activity that is both spatially and temporally sensitive [8, 39]. While our findings suggest that slope increases precede seizures, these changes should not be interpreted as a direct and exclusive marker of inhibition. Instead, they may reflect a broader reduction in network excitability involving multiple mechanisms, including synaptic filtering and connectivity sparsity. Nonetheless, our findings show a consistent pre-ictal increase in the exponent, suggesting that it could form the basis of a predictive feature. In fact, the aperiodic exponent can be continuously tracked using scalp EEG, representing a sensitive feature with potential application in seizure prediction. Future development of closed-loop systems could leverage this metric for early seizure detection, potentially enabling adaptive neuromodulation to prevent seizures before clinical onset. However, this interpretation requires careful validation: at present, these results should be regarded as candidate markers rather than established predictors. In fact, while our findings are compelling, there are limitations to be acknowledged. First, the study used a modest-sized retrospective cohort, with heterogeneity in seizure characteristics. Second, although sleep/wake state is known to influence the aperiodic exponent, the majority of seizures in our dataset (75%) occurred during wakefulness,

precluding a reliable statistical comparison across vigilance states. Finally, the retrospective design precludes firm conclusions about predictive validity; therefore, the observed pre-ictal dynamics should be interpreted as correlates rather than prospective predictors of seizure onset. Future work employing prospective validation frameworks will be essential to establish their forecasting value [24, 70]. Nevertheless, the convergence of dynamic, anatomical, and neurochemical data provides a robust framework for future studies. Ongoing work should aim to replicate these findings in larger cohorts and evaluate their predictive value in prospective seizure forecasting pipelines. Integration with multimodal imaging and real-time EEG analytics may enable the development of personalized seizure prediction tools grounded in E/I balance dynamics.

Conclusions

In summary, our findings suggest a temporally dynamic and spatially widespread shift toward cortical inhibition at the whole-brain scale in the moments preceding seizure onset. These changes appear related to patterns of functional connectivity, regional cortical structure, and neurotransmitter receptor architecture. Together, these observations highlight the importance of considering both local and global neural dynamics in understanding seizure susceptibility, while emphasizing the need for cautious interpretation and further validation.

Abbreviations

EEG	Electroencephalography
E/I	Excitation/Inhibition balance
sEEG	Stereo electroencephalography
SLE	Seizure-like event
ASM	Antiseizure medication
PSD	Power spectrum density
FDR	False discovery rate
RS	Resting state
PET	Positron emission tomography
GC	Granger causality
ROI	Region of interest
STFT	Short-time Fourier transform
MRI	Magnetic resonance imaging
BEM	Boundary element method
HDEEG	High-density electroencephalography
ICA	Independent component analysis
FCD	Focal cortical dysplasia
HS	Hippocampal sclerosis
ACNS	American Clinical Neurophysiology Society

Supplementary Information

The online version contains supplementary material available at <https://doi.org/10.1186/s12916-025-04447-7>.

Supplementary Material 1. Figures S1-S2. Fig.S1-[Additional pre-ictal time interval results]. Fig.S2-[Effective connectivity in theta band and its relationship with the aperiodic exponent].

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Authors' contributions

GMD: conceptualization, methodology, software, data curation, formal analysis, visualization, writing—original draft, project administration; SC: data curation, formal analysis, visualization; AD: conceptualization, writing original draft; JYH: data curation, methodology, writing—review and editing; LA: data curation, resources; VV: data curation, resources; EO: data curation, resources; PB: conceptualization, supervision, writing—review and editing; GP: conceptualization, methodology, supervision, writing—original draft. All authors read and approved the final manuscript.

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Data availability

The data that support the findings of this study are available on request to the corresponding author. The raw data are not publicly available due to privacy or ethical restrictions. All the scripts are available at the following GitHub page: <https://github.com/simone-cuozzo/dynamic-fooof-prior-seizure>. Brain maps of the receptor density of neurotransmitters used in this manuscript are publicly available: Network Neuroscience Lab Github: https://github.com/netneurolab/hansen_receptors/tree/main/data/PET_parcellated/scale033.

Declarations

Ethics approval and consent to participate

The study protocol (Protocol number: 1018 (cod. int. CET ANV 2023–112)) was conducted according to the Declaration of Helsinki and approved by the local ethical committee (Comitato Etico Area Nord Veneto). All participants provided written informed consent to participate in the study.

Consent for publication

All authors have agreed to the submission and publication of this research.

Competing interests

The authors declare no competing interests.

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References

- Atallah BV, Scanziani M. Instantaneous modulation of gamma oscillation frequency by balancing excitation with inhibition. *Neuron*. 2009;62(4):566–77.
- Zhou S, Yu Y. Synaptic E-I balance underlies efficient neural coding. *Front Neurosci*. 2018;12:46.
- Lauterborn JC, Scaduto P, Cox CD, Schulmann A, Lynch G, Gall CM, et al. Increased excitatory to inhibitory synaptic ratio in parietal cortex samples from individuals with Alzheimer's disease. *Nat Commun*. 2021;12(1):2603.
- McKeown DJ, Jones M, Pihl C, Finley AJ, Kelley N, Baumann O, et al. Medication-invariant resting aperiodic and periodic neural activity in Parkinson's disease. *Psychophysiology*. 2024;61(4):e14478.
- van van Hugte EJJ, Schubert D, Nadif Kasri N. Excitatory/inhibitory balance in epilepsies and neurodevelopmental disorders: depolarizing γ -aminobutyric acid as a common mechanism. *Epilepsia*. 2023;64(8):1975–90.
- Žiburkus J, Cressman JR, Schiff SJ. Seizures as imbalanced up states: excitatory and inhibitory conductances during seizure-like events. *J Neurophysiol*. 2013;109(5):1296–306.
- Ahmad J, Ellis C, Leech R, Voytek B, Garces P, Jones E, et al. From mechanisms to markers: novel noninvasive EEG proxy markers of the neural excitation and inhibition system in humans. *Transl Psychiatry*. 2022;12(1):1–12.
- Donoghue T, Haller M, Peterson EJ, Varma P, Sebastian P, Gao R, et al. Parameterizing neural power spectra into periodic and aperiodic components. *Nat Neurosci*. 2020;23(12):1655–65.
- Gao R, Peterson EJ, Voytek B. Inferring synaptic excitation/inhibition balance from field potentials. *Neuroimage*. 2017;158:70–8.
- Gil Avila C, May ES, Bott FS, Tiemann L, Hohn V, Heitmann H, et al. Assessing the balance between excitation and inhibition in chronic pain through the aperiodic component of EEG. *Elife*. 2025;13:RP101727.
- Martínez-Cañada P, Perez-Valero E, Minguillon J, Pelayo F, López-Gordo MA, Morillas C. Combining aperiodic 1/f slopes and brain simulation: an EEG/MEG proxy marker of excitation/inhibition imbalance in Alzheimer's disease. *Alzheimer's Dementia*. 2023;15(3):e12477.
- McKeon SD, Perica MI, Parr AC, Calabro FJ, Foran W, Hetherington H, et al. Aperiodic EEG and 7T MRSI evidence for maturation of E/I balance supporting the development of working memory through adolescence. *Dev Cogn Neurosci*. 2024;66:101373.
- Stanyard RA, Mason D, Ellis C, Dickson H, Short R, Batalle D, et al. Aperiodic and Hurst EEG exponents across early human brain development: a systematic review. *Dev Cogn Neurosci*. 2024;68:101402.
- Schuler AL, Tik M, Kallioniemi E, Suller Marti A, Cai Z, Pellegrino G. Approaches to map cortical excitability beyond the primary motor cortex—perspectives from cognitive neuroscience, multimodal imaging and clinical applications. *Neurosci Biobehav Rev*. 2025;177:106338.
- Gerster M, Waterstraat G, Litvak V, Lehnertz K, Schnitzler A, Florin E, et al. Separating neural oscillations from aperiodic 1/f activity: challenges and recommendations. *Neuroinformatics*. 2022;20(4):991–1012.
- Salvatore SV, Lambert PM, Benz A, Rensing NR, Wong M, Zorumski CF, et al. Periodic and aperiodic changes to cortical EEG in response to pharmacological manipulation. *J Neurophysiol*. 2024;131(3):529–40.
- Sheldon AA, Moser HR, Abdullahi KS, Allison KD, Mulder CB, Montoya SA, et al. Aperiodic slope reflects glutamatergic tone in the human brain. *bioRxiv*; 2025;p. 2025.05.15.654118.
- Donoghue T. A systematic review of aperiodic neural activity in clinical investigations. *medRxiv*; 2025; p. 2024.10.14.24314925.
- Duma GM, Cuozzo S, Wilson L, Danielli A, Bonanni P, Pellegrino G. Excitation/inhibition balance relates to cognitive function and gene expression in temporal lobe epilepsy: a high density EEG assessment with aperiodic exponent. *Brain Commun*. 2024;6(4):fcae231.
- Pellegrino G, Hedrich T, Sziklas V, Lina JM, Grova C, Kobayashi E. How cerebral cortex protects itself from interictal spikes: the alpha/beta inhibition mechanism. *Hum Brain Mapp*. 2021;42(11):3352–65.
- Doss DJ, Shless JS, Bick SK, Makhoul GS, Negi AS, Bibro CE, et al. The interictal suppression hypothesis is the dominant differentiator of seizure onset zones in focal epilepsy. *Brain*. 2024;147(9):3009–17.
- Jirsa VK, Stacey WC, Quilichini PP, Ivanov AI, Bernard C. On the nature of seizure dynamics. *Brain*. 2014;137(8):2210–30.
- Luo Z, Peng K, Liang Z, Cai S, Xu C, Li D, et al. Mapping effective connectivity by virtually perturbing a surrogate brain. *Nat Methods*. 2025. <https://doi.org/10.1038/s41592-025-02654-x>.
- Jirsa VK, Proix T, Perdakis D, Woodman MM, Wang H, Gonzalez-Martinez J, et al. The virtual epileptic patient: individualized whole-brain models of epilepsy spread. *Neuroimage*. 2017;145:377–88.
- Pellegrino G, Hedrich T, Chowdhury R, Hall JA, Lina JM, Dubeau F, et al. Source localization of the seizure onset zone from ictal EEG/MEG data. *Hum Brain Mapp*. 2016;37(7):2528–46.

26. Satzer D, Kaye LC, Ojemann SG, Kramer DR, Thompson JA. Aperiodic activity as a biomarker of seizures and neuromodulation. *Brain Stimulat.* 2025;18(3):738–44.
27. Hansen JY, Shafiei G, Markello RD, Smart K, Cox SML, Nørgaard M, et al. Mapping neurotransmitter systems to the structural and functional organization of the human neocortex. *Nat Neurosci.* 2022;25(11):1569–81.
28. Shafiei G, Fulcher BD, Voytek B, Satterthwaite TD, Baillet S, Mísic B. Neurophysiological signatures of cortical micro-architecture. *Nat Commun.* 2023;14(1):6000.
29. Duma GM, Pellegrino G, Rabuffo G, Danielli A, Antoniazzi L, Vitale V, et al. Altered spread of waves of activities at large scale is influenced by cortical thickness organization in temporal lobe epilepsy: a MRI - high-density EEG study. *Brain Commun.* 2023;fcad348.
30. Kwan P, Arzimanoglou A, Berg AT, Brodie MJ, Allen Hauser W, Mathern G, et al. Definition of drug resistant epilepsy: consensus proposal by the ad hoc task force of the ILAE commission on therapeutic strategies. *Epilepsia.* 2010;51(6):1069–77.
31. Hirsch LJ, Fong MWK, Leutinger M, LaRoche SM, Beniczky S, Abend NS, et al. American clinical neurophysiology society's standardized critical care EEG terminology: 2021 version. *J Clin Neurophysiol.* 2021;38(1):1–29.
32. Pion-Tonachini L, Kreutz-Delgado K, Makeig S. ICLABEL: an automated electroencephalographic independent component classifier, dataset, and website. *Neuroimage.* 2019;198:181–97.
33. Fischl B, Dale AM. Measuring the thickness of the human cerebral cortex from magnetic resonance images. *Proc Natl Acad Sci U S A.* 2000;97(20):11050–5.
34. Tadel F, Baillet S, Mosher JC, Pantazis D, Leahy RM. Brainstorm: a user-friendly application for MEG/EEG analysis. *Comput Intell Neurosci.* 2011;2011:1–13.
35. Gramfort A, Papadopoulos T, Olivi E, Clerc M. OpenMEEG: open source software for quasistatic bioelectromagnetics. *Biomed Eng Online.* 2010;9(1):45.
36. Hämäläinen MS, Ilmoniemi RJ. Interpreting magnetic fields of the brain: minimum norm estimates. *Med Biol Eng Comput.* 1994;32(1):35–42.
37. Desikan RS, Ségonne F, Fischl B, Quinn BT, Dickerson BC, Blacker D, et al. An automated labeling system for subdividing the human cerebral cortex on MRI scans into gyral based regions of interest. *Neuroimage.* 2006;31(3):968–80.
38. Duma GM, Danielli A, Vettorel A, Antoniazzi L, Mento G, Bonanni P. Investigation of dynamic functional connectivity of the source reconstructed epileptiform discharges in focal epilepsy: a graph theory approach. *Epilepsy Res.* 2021;176:106745.
39. Wilson LE, da Silva Castanheira J, Baillet S. Time-resolved parameterization of aperiodic and periodic brain activity. Jensen O, de Lange FP, van Es M, Kujala J, editors. *eLife.* 2022;11:e77348.
40. Oostenveld R, Fries P, Maris E, Schoffelen JM. FieldTrip: open source software for advanced analysis of MEG, EEG, and invasive electrophysiological data. *Comput Intell Neurosci.* 2011;2011:1–19.
41. Zamani Esfahani F, Jo Y, Faskowitz J, Byrge L, Kennedy DP, Sporns O, et al. High-amplitude co-fluctuations in cortical activity drive functional connectivity. *Proc Natl Acad Sci U S A.* 2020;117(45):28393–401.
42. Del Popolo Cristaldi F, Buodo G, Duma GM, Sarlo M, Mento G. Unbalanced functional connectivity at rest affects the ERP correlates of affective prediction in high intolerance of uncertainty individuals: a high density EEG investigation. *Int J Psychophysiol.* 2022;178:22–33.
43. van den Heuvel MP, de Lange SC, Zalesky A, Seguin C, Yeo BTT, Schmidt R. Proportional thresholding in resting-state fMRI functional connectivity networks and consequences for patient-control connectome studies: issues and recommendations. *Neuroimage.* 2017;152:437–49.
44. Treiman DM. GABAergic mechanisms in epilepsy. *Epilepsia.* 2001;42(Suppl 3):8–12.
45. Wang Y, Tan B, Wang Y, Chen Z. Cholinergic signaling, neural excitability, and epilepsy. *Molecules.* 2021;26(8):2258.
46. Chen TS, Huang TH, Lai MC, Huang CW. The role of glutamate receptors in epilepsy. *Biomedicines.* 2023;11(3):783.
47. Hammers A, Asselin MC, Hinz R, Kitchen I, Brooks DJ, Duncan JS, et al. Upregulation of opioid receptor binding following spontaneous epileptic seizures. *Brain.* 2007;130(4):1009–16.
48. Sourbron J, Lagae L. Serotonin receptors in epilepsy: novel treatment targets? *Epilepsia Open.* 2022;7(2):231–46.
49. Larivière S, Paquola C, Park B yong, Royer J, Wang Y, Benkarim O, et al. The ENIGMA Toolbox: multiscale neural contextualization of multisite neuroimaging datasets. *Nat Methods.* 2021;18(7):698–700.
50. Alexander-Bloch AF, Shou H, Liu S, Satterthwaite TD, Glahn DC, Shinohara RT, et al. On testing for spatial correspondence between maps of human brain structure and function. *Neuroimage.* 2018;178:540–51.
51. Tóth K, Hofer KT, Kandrás Á, Entz L, Bagó A, Erőss L, et al. Hyperexcitability of the network contributes to synchronization processes in the human epileptic neocortex. *J Physiol.* 2018;596(2):317–42.
52. Sills GJ, Rogawski MA. Mechanisms of action of currently used antiseizure drugs. *Neuropharmacology.* 2020;168:107966.
53. Shao LR, Habela CW, Stafstrom CE. Pediatric epilepsy mechanisms: expanding the paradigm of excitation/inhibition imbalance. *Children.* 2019;6(2):23.
54. Englot DJ, Yang L, Hamid H, Danielson N, Bai X, Marfeo A, et al. Impaired consciousness in temporal lobe seizures: role of cortical slow activity. *Brain.* 2010;133(12):3764–77.
55. Sumsy S, Greenfield LJ. Network analysis of preictal iEEG reveals changes in network structure preceding seizure onset. *Sci Rep.* 2022;12(1):12526.
56. Makhoul GS, Doss DJ, Johnson GW, Withers CP, Cavender AC, Hidalgo Monroy Lerma B, et al. Collapse of interictal suppressive networks permits seizure spread. *Brain.* 2025;awaf215.
57. Narasimhan S, Kundassery KB, Gupta K, Johnson GW, Wills KE, Goodale SE, et al. Seizure-onset regions demonstrate high inward directed connectivity during resting-state: an SEEG study in focal epilepsy. *Epilepsia.* 2020;61(11):2534–44.
58. Aru J, Aru J, Priesemann V, Wibral M, Lana L, Pipa G, et al. Untangling cross-frequency coupling in neuroscience. *Curr Opin Neurobiol.* 2015;31:51–61.
59. Buzsáki G, Draguhn A. Neuronal oscillations in cortical networks. *Science.* 2004;304(5679):1926–9.
60. Faust TE, Devlin BA, Farhy-Tselnicker I, Ferro A, Postolache M, Xin W. Glial control of cortical neuronal circuit maturation and plasticity. *J Neurosci.* 2024;44(40):e1208242024.
61. Proskovec AL, Spooner RK, Wiesman AI, Wilson TW. Local cortical thickness predicts somatosensory gamma oscillations and sensory gating: a multimodal approach. *Neuroimage.* 2020;214:116749.
62. Tatti R, Haley MS, Swanson OK, Tselha T, Maffei A. Neurophysiology and regulation of the balance between excitation and inhibition in neocortical circuits. *Biol Psychiatry.* 2017;81(10):821–31.
63. Pagonabarraga J, Corcuera-Solano I, Vives-Gilabert Y, Llebaria G, García-Sánchez C, Pascual-Sedano B, et al. Pattern of regional cortical thinning associated with cognitive deterioration in Parkinson's disease. *PLoS ONE.* 2013;8(1):e54980.
64. Monteverdi A, Palesi F, Costa A, Vitali P, Pichiechio A, Cotta Ramusino M, et al. Subject-specific features of excitation/inhibition profiles in neurodegenerative diseases. *Front Aging Neurosci.* 2022;14:868342.
65. Liu X, Bautista J, Liu E, Zikopoulos B. Imbalance of laminar-specific excitatory and inhibitory circuits of the orbitofrontal cortex in autism. *Mol Autism.* 2020;11:83.
66. Galovic M, Van Dooren VQH, Postma TS, Vos SB, Caciagli L, Borzi G, et al. Progressive cortical thinning in patients with focal epilepsy. *JAMA Neurol.* 2019;76(10):1230.
67. Larivière S, Rodríguez-Cruces R, Royer J, Caligiuri ME, Gambardella A, Concha L, et al. Network-based atrophy modeling in the common epilepsies: a worldwide ENIGMA study. *Sci Adv.* 2020;6(47):eabc6457.
68. Akyuz E, Polat AK, Eroglu E, Kullu I, Angelopoulou E, Paudel YN. Revisiting the role of neurotransmitters in epilepsy: an updated review. *Life Sci.* 2021;265:118826.
69. Becchetti A, Grandi LC, Cerina M, Amadeo A. Nicotinic acetylcholine receptors and epilepsy. *Pharmacol Res.* 2023;189:106698.
70. Makhlova J, Medina Villalon S, Wang H, Giusiano B, Woodman M, Bénar C, et al. Virtual epileptic patient brain modeling: relationships with seizure onset and surgical outcome. *Epilepsia.* 2022;63(8):1942–55.

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